

Project Plan

The GZigZag project studies and puts into practice information management methods more advanced than those in use today, building on Ted Nelson's ZigZag and Xanadu models. The starting point is that current information management methods are inadequate for meeting the challenges of the future. The group's essential areas of research are user interfaces and the storage of information in hyperstructures, as well as applying the results of this research in practice.

Managing large amounts of information is very difficult with current systems, because in a file system the only information between files is the directory structure. A directory structure is a hierarchy, and is therefore suited only to a simple division into categories. In HTML the shortcomings of the directory structure have been patched with links that are stored inside the file. These links are one-directional and break easily.

Databases are a much more advanced way of managing information than file systems. They are, however, very rigid systems: a database is usually designed for the use of a single application, and is not suited to storing general information; database systems offer no more possibilities for linking between databases than file systems do; and databases are seldom accessible from the outside—often only a WWW interface is available. The most significant advantages of databases over file systems are the possibilities of efficient information retrieval and of storing the relationships between pieces of information, but they are not flexible enough for general information management.

The main goal of the project is to create the GZigZag software, on top of which programs that are better than today's in their presentation, search and usage features can easily be built. The software is based on the ZigZag structure, which consists of cells and of connections made along different dimensions. In the structure, information is held in cells, each of which contains a much smaller part of the information than current files do; one cell contains roughly as much information as a spreadsheet cell. Unlike in a spreadsheet, the cells are connected to one another in a multidimensional, flexible structure in which every cell can be connected to any other cells. The dimensions of the structure can have different meanings, such as "the contents of minutes 3141S", "reference", "elaboration for those interested in the matter", or "directory structure". Nor do the dimensions have the same restrictions as the real world: moving first up and then right can lead to a different place than moving first right and then up.

The bidirectionality of connections is one of the most important properties of the ZigZag structure. On the WWW, which uses one-directional links, it is not possible, for example, to add one's own marginal notes to pages, or to browse the pages that refer to or quote a passage of a document. In the ZigZag system these things are easy,

and the filtering made possible by auxiliary information also prevents the matter at hand from drowning in a jungle of various marginal notes and elaborations.

In developing the GZigZag software, the aim is to study the benefits of various visual methods when they are used throughout the whole system. The group seeks to improve the user interfaces of applications by abandoning the traditional “windows, menus and buttons” model: information must be presented in its context, not chopped into boxes that have no connections with one another. Perhaps the most important of the methods is Focus+Context, whose main idea is to bring the information the user is interested in at any given moment (Focus) to the fore as prominently as possible, and the related things (Context) according to how important they are with respect to the focus. Closely related to Focus+Context is animation: when the user moves around in the information structure, it is very important to show how the new information relates to the previous, and the best way to show this is animation.

The group writes articles on the results of its research for international conferences and publications, including UIST, HICSS and HT01. Articles are currently being prepared on applying Focus+Context to message threads and to large collections of text documents, as well as an article presenting the ZigZag model at a general level.

The development of the GZigZag software is public, and, like the successful Linux and Apache, the software is distributed under a free license (GNU LGPL). At present the implementation is at the prototype stage, and it is available over the network at <http://www.gzigzag.org/>.

The following goals have been set for the end of 2001:

- A prototype of a messaging, calendar, etc. system (Sonera).
- A prototype of a new kind of word processor (Nile). Nile differs from current programs, among other things, in that text can be quoted so that the document the quotation comes from is shown next to the quotation. Things related to the document, such as pictures, float beside the text, so that a picture related to even a long text stays in view the whole time, even when scrolling forward through the text. Versioning ensures that old versions can always be read, and quotation links (transclusions) ensure that intertextual connections can be presented to the user in an understandable way.

By the end of 2002 the aim is to implement the GZigZag software in cell languages (“bootstrap”). This first requires developing cell languages suited to the task, and is one of the most important phases of the development work. The move to cell languages makes possible the quick writing of clear program code that handles the ZigZag structure, and therefore this phase is very essential both for the research work and for the development of the GZigZag software.

The person responsible for the project is Tuomas J. Lukka, PhD. Tasks have been divided among the project's staff as follows:

Katariina Ervasti	Research on humanistic aspects, documentation and popularization.
Tuukka Hastrup	ZigZag theory.
Antti-Juhani Kaijanaho	Version control of cell spaces, computer networks, cell languages and ZigZag theory.
Tuomas Lukka	Project management and control of the whole.
Tero Mäyränen	Use of ZigZag in multimedia applications, and cell languages.
Vesa Parkkinen	Utilizing ZigZag in palmtop computers, and multimedia.
Rauli Ruohonen	Cell languages, ZigZag theory and palmtop computers.
Kimmo Wideroos	Computer-assisted learning and process writing.

The group cooperates with Professor Ted Nelson (Keio University, Japan; Xanadu, Inc.) and the German student Benjamin Fallenstein. Professor Nelson, together with Dr. Lukka, develops the ZigZag and Xanadu models further, and Fallenstein is involved in the development work on the GZigZag software.

GZigZag and its basic ideas can in principle be utilized in producing any software aimed at end users, but it is especially well suited to systems that operate in computer networks and to integrated systems.

Risks associated with the project include:

- ZigZag remaining little used, because it differs so much from current systems that the threshold for moving to it may be high.
- The spread of closed data storage methods may create obstacles to the adoption of ZigZag, since they limit the possibility of making GZigZag compatible with other systems. Files created by programs that use closed storage methods cannot be read with other programs.
- Difficulty of implementation.
- Key people leaving the project.